

BASIC ANTIVIRUS SIMULATION USING PYTHON

Submitted By

Name: Shreenidhi Nandana

Internship: CyberSecurity Internship

1. Introduction

Antivirus software is an essential security tool used to detect and prevent malicious files from affecting computer systems. Most traditional antivirus solutions use signature-based detection methods, where file signatures are compared against a database of known malware signatures.

This project implements a Basic Antivirus Simulation using Python. The application scans files, generates SHA-256 hashes, compares them against a malware signature database, and quarantines detected malicious files.

2. Objective

The main objective of this project is to develop a simple antivirus simulation capable of:

- Scanning files in a directory.
 - Generating SHA-256 hashes for files.
 - Comparing file hashes with known malware signatures.
 - Detecting malicious files.
 - Moving detected files to a quarantine folder.
-

3. Technologies Used

- Python
 - hashlib Module
 - os Module
 - shutil Module
-

4. Project Workflow

The antivirus simulation follows the steps below:

1. Select files from the target directory.
 2. Generate SHA-256 hash values.
 3. Load malware signatures from the database.
 4. Compare generated hashes with stored signatures.
 5. Detect malicious files.
 6. Move malicious files to the quarantine folder.
 7. Display scan results to the user.
-

5. Implementation

The project uses SHA-256 hashing to create unique fingerprints for files. The generated hash is compared with the malware signature database stored in the signatures file.

If a match is found, the file is identified as malicious and automatically moved to the quarantine folder. Files that do not match any signature are marked as safe.

6. Results

The antivirus simulation successfully detected malicious files and isolated them from the scanning directory.

Sample Output:

```
PS E:\Antivirus Simulator> python antivirus.py
[!] Malware Found: malware.txt
Moved malware.txt to quarantine
[✓] Safe: safe.txt
PS E:\Antivirus Simulator
```

7. Conclusion

The Basic Antivirus Simulation successfully demonstrates the fundamental concepts of signature-based malware detection. Through hashing, signature comparison, and quarantine mechanisms, the project provides practical knowledge of how antivirus software identifies and isolates malicious files.

8. Future Enhancements

- Graphical User Interface (GUI)
- Real-Time File Monitoring
- Automatic Log Generation
- Cloud-Based Signature Updates
- VirusTotal API Integration.